Rudraksh Bhandari

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Education

University of California San Diego — San Diego, CA
Bachelor of Science in Computer Science | GPA: **3.9** — *Expected March 2027*

Experience

Amazon Web Services - EC2GN (Hyperplane) — Santa Clara, CA
Software Development Engineer Intern — *June 2026 – September 2026*

- Built a Python gameday fault-injection framework for AWS Hyperplane, a hyperscale network virtualization platform, translating **3 historical customer-impacting incidents** into repeatable failure scenarios using AWS Fault Injection Service and service APIs.
- Designed TCP peer-exchange and UDP health-probe failure scenarios across a pre-production Hyperplane fleet of **2,500+ hosts** by tracing failure propagation through distributed host and target health systems, validating downstream behavior via S3 configuration files and DynamoDB health records.
- Reduced a multi-step manual gameday workflow to a single CLI command, automating fault injection, CloudWatch evidence collection, health validation, rollback, and recovery.

Software Development Engineer Intern — *June 2025 – September 2025*

- Built a React/TypeScript CLI command generator for on-call incident response, cutting manual engineer effort by **75%** and reducing command preparation time from **over 2 minutes to under 30 seconds**.
- Engineered a Java/Python AWS event service using Lambda, SSM, SQS, and SNS to surface patch status across EC2 cluster groups, improving visibility for root-cause analysis during incidents.

Outfitr — San Diego, CA
Co-founder & Lead Developer — *April 2026 – Present*

- Launched a fashion app that helps Gen Z decide what to wear from clothes they already own, with closet import, outfit recommendations, and friend polls; ranked in the **top 10% of applicants** for **Y Combinator** Summer and Fall '26.
- Architected a production multimodal closet-extraction pipeline using SAM3, VLM-based classification, and embedding-based deduplication, processing **200+ real-customer photos across 100+ jobs** and adding **300+ garments** to users' closets.

ShareAllBooks — United Arab Emirates
Founder — *August 2023 – September 2024*

- Built and launched a React Native book-sharing app on App Store & Play Store, **reaching 2,000+ downloads and 1,500+ donations/sales** within months; featured in UAE national press; peaked at **#41 in App Store's Top Charts** for Books.

Projects & Research

NyaayWatch — *April 2026 – Present*

- Built and operate an open-source court-data platform serving **10,000+ monthly visitors**, publishing versioned data across India's Supreme Court, all **25 High Courts**, and **36 lower-court geographies** through public APIs, evidence packs, and interactive dashboards.
- Designed a reproducible data pipeline that separates raw court-data captures from published data, storing evidence in S3 and immutable snapshots in Postgres with append-only publication history for validation, replay, and rollback.

Undergraduate Research — *March 2026 – July 2026*

- Developing a unified model for generation, embeddings, and agentic routing in the Statistical Visual Computing Lab (SVCL) under Prof. Vasconcelos, using LoRA-based embedding distillation and generation replay to improve retrieval.
- Benchmarking dense prefill vs. KVCOMM TTFT on H100s across multi-agent workloads in the Alternative Computing Technologies (ACT) Lab under Prof. Esmaeilzadeh to evaluate KV-cache reuse.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, C++
Databases: PostgreSQL, Redis, DynamoDB, MongoDB
Frameworks: React Native, React, Next.js, Node, Express
Tools: Git, AWS, Docker, Cloudflare, Xcode, Expo